

PostgreSQL - GROUPING SETS, ROLLUP & CUBE (Complete Notes)

Practice Dataset

```
CREATE TABLE sales_demo (  
    id INT PRIMARY KEY,  
    continent VARCHAR(30),  
    country VARCHAR(30),  
    city VARCHAR(30),  
    amount INT  
);  
  
INSERT INTO sales_demo VALUES  
(1, 'Asia', 'India', 'Bangalore', 1000),  
(2, 'Asia', 'India', 'Chennai', 2000),  
(3, 'Asia', 'Japan', 'Tokyo', 4000),  
(4, 'Asia', 'Japan', 'Hiroshima', 5000),  
(5, 'Europe', 'United Kingdom', 'London', 1000),  
(6, 'Europe', 'United Kingdom', 'Manchester', 2000),  
(7, 'Europe', 'France', 'Paris', 4000),  
(8, 'Europe', 'France', 'Cannes', 5000);
```

Cheat Sheet

1. Normal GROUP BY

```
By Continent  
SELECT continent, SUM(amount)  
FROM sales_demo  
GROUP BY continent;
```

```
By Country  
SELECT country, SUM(amount)  
FROM sales_demo  
GROUP BY country;
```

```
By City  
SELECT city, SUM(amount)  
FROM sales_demo  
GROUP BY city;
```

2. GROUPING SETS

Think:
"I choose exactly what summaries I want."

Country + Grand Total

```
SELECT continent, country, SUM(amount)  
FROM sales_demo  
GROUP BY GROUPING SETS  
(  
    (country),  
    ()  
);
```

Continent + Country

```
SELECT continent, country, SUM(amount)  
FROM sales_demo  
GROUP BY GROUPING SETS  
(  
    (continent),  
    (country)  
);
```

Country + City + Grand Total

```
SELECT country,city,SUM(amount)
FROM sales_demo
GROUP BY GROUPING SETS
(
(country),
(city),
()
);
```

Everything you choose

```
SELECT continent,country,city,SUM(amount)
FROM sales_demo
GROUP BY GROUPING SETS
(
(continent),
(country),
(city),
()
);
```

3. ROLLUP

Think:
"Go from detailed to summarized."

```
SELECT continent,country,SUM(amount)
FROM sales_demo
GROUP BY ROLLUP(continent,country);
```

Equivalent to

```
GROUPING SETS
(
(continent,country),
(continent),
()
)
```

Continent -> Country -> City -> Total

```
SELECT continent,country,city,SUM(amount)
FROM sales_demo
GROUP BY ROLLUP(continent,country,city);
```

Equivalent to

```
(continent,country,city)
(continent,country)
(continent)
()
```

Notice:
ROLLUP removes one column at a time from RIGHT to LEFT.

4. CUBE

Think:
"Give me every possible subtotal."

```
SELECT continent,country,SUM(amount)
FROM sales_demo
GROUP BY CUBE(continent,country);
```

Equivalent to

```
(continent,country)
(continent)
(country)
()
```

Three columns

```
SELECT continent, country, city, SUM(amount)
FROM sales_demo
GROUP BY CUBE(continent, country, city);
```

Equivalent to

```
(continent, country, city)
(continent, country)
(continent, city)
(country, city)
(continent)
(country)
(city)
()
```

There are $2^3 = 8$ grouping sets.

Practice Questions

Q1 Show total sales by continent.

Q2 Show total sales by country.

Q3 Show Continent totals, Country totals and Grand Total using GROUPING SETS.

Q4 Show Continent, Country, City, Country Total, Continent Total and Grand Total using ROLLUP.

Q5 Show every possible subtotal using CUBE.

Q6 Rewrite ROLLUP using GROUPING SETS.

Q7 Rewrite CUBE using GROUPING SETS.

One-Page Notes

GROUPING SETS

You choose the summaries.

GROUPING SETS

```
(
(A),
(B),
(C),
()
)
```

Equivalent to multiple GROUP BY queries combined with UNION ALL.

ROLLUP

Hierarchy.

ROLLUP(A, B, C)

Equivalent to

```
(A, B, C)
(A, B)
(A)
()
```

Removes columns from RIGHT to LEFT.

CUBE

Every possible combination.

CUBE(A, B, C)

Equivalent to

(A, B, C)
(A, B)
(A, C)
(B, C)
(A)
(B)
(C)
()

Number of grouping sets = 2^n

Relationship

ROLLUP $\subset$ GROUPING SETS

CUBE $\subset$ GROUPING SETS

GROUPING SETS is the most flexible.
ROLLUP and CUBE are shortcuts.